Jiefu (Jeff) Zhang

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Research focus: AI for chip design and manufacturing automation (EDA)

Education

Purdue University — West Lafayette, IN
M.S. Computer Engineering (Thesis Track) — *Expected Dec 2026*
B.S. Computer Engineering — *Dec 2024*

- Accelerated BS/MS Program | Graduate GPA: 3.88/4.0
- Coursework: ASIC Design, VLSI Design, Semiconductor Fabrication, RISC-V Architecture, Embedded Systems, Artificial Intelligence

Research Experience

Graduate Research Assistant, CLAN Lab, Purdue University — Oct 2024 – Present
MS Student Member, Wan Lab, Columbia University — Jun 2026 – Present
Advisors: Prof. Vaneet Aggarwal (Purdue) and Prof. Zishen Wan (Columbia)

- **CARVE: Chip Placement with Reusable Expertise** (first author, arXiv:2609.33106): Adapted pre-trained RL macro-placement policies to new circuits without the original training data, reusing saved specialists only after they pass local validation. A reuse-first variant cut recorded training time by **58.5%** (9.66 → 4.01 h) while mean HPWL gain moved only from 8.41% to 7.86%; imported specialists were reused on **6 of 7** new IBM circuits.
- **SIREN: Correcting the Winner's Curse in LLM Evaluation** (first author, **NeurIPS 2026**, arXiv): Co-developed a selection-aware repeated-split reporting protocol with bootstrap confidence bands; led MMLU-Pro tuning experiments with vLLM-served open-weight models and DSPy. Now extending it to AI-for-EDA results (macro placement).
- **Safe Social Robot Navigation** (first author, arXiv:2603.06729, code): Built a PPO-based navigation framework with density-adaptive proxemic reward shaping; achieved **86%+ safe success at 21-agent crowd densities** where DS-RNN and SARL baselines collapse, benchmarked against ORCA and SFM.
- **Continual and Decentralized Learning** (SPHERE, Core-Halo): Co-developed representation-guided replay allocation for online continual learning; built SUMO/PettingZoo multi-intersection traffic-control benchmarks and a decentralized DQN pipeline for decentralized fixed-point learning.

Selected Hardware Projects

- **Multicore RISC-V Processor:** Designed a 5-stage pipelined RISC-V processor with hazard resolution via forwarding/stalling; improved multicore efficiency by **30%** using MOESI cache coherence.
- **USB 1.1 ASIC Data Transfer:** Achieved **12 Mbps** throughput (USB 1.1 standard) in Verilog with AHB-Lite interface; validated via QuestaSim with **98% test coverage**.
- **8-bit Wallace Tree Multiplier** (ECE 559 MOS VLSI Design): Full-custom schematic/layout, DRC/LVS clean; worst-case delay $\leq$ **2.5 ns**, energy $\leq$ 850 fJ with parasitic extraction.

Technical Skills

- **Programming:** Python, C/C++, Verilog, SystemVerilog, Java, Bash, Git, LaTeX
- **ML / AI for EDA:** PyTorch, RL (PPO, DQN), RL-based macro placement, continual learning, LLM evaluation, DSPy, vLLM, PettingZoo, Multi-GPU Training
- **Hardware/EDA:** Cadence Virtuoso, QuestaSim, DRC/LVS, parasitic extraction, RTL Design, FPGA, ASIC/SoC Integration, Linux

Additional

- **Award:** NeurIPS 2026 Financial Assistance Award (full conference and workshop registration and hotel stay)
- **Teaching:** Graduate TA, Purdue ECE 33700 – ASIC Design Laboratory, 60+ students (Spring 2025)
- **Industry Intern:** Product Maintenance Engineer Intern, Compound Co., Shenzhen (Summer 2023) – sensor calibration & IoT for medical devices
- **Languages:** English (fluent, in the US since 2016), Mandarin (native)